

# Two Thirds of the Vertices of a Minimal Counterexample to the Erdős–Gyárfás Conjecture are Cubic

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## Abstract

A minimal counterexample to the Erdős–Gyárfás conjecture is a graph of minimum possible order and, subject to that, minimum possible size, with minimum degree at least 3 and containing no cycle whose length is a power of 2. Carr recently showed that at least  $4/7$  of the vertices of any minimal counterexample have degree exactly 3. We observe that combining two structural facts already established in that work yields the stronger bound  $2/3$ , and we prove an additional structural restriction: whenever the four edges leaving a pair of adjacent cubic vertices all end in vertices of degree at least 4, the pair has a unique common neighbor, and that common neighbor has degree exactly 4.

**Keywords:** Erdős–Gyárfás conjecture, cycles of power-of-two length, cubic graphs, minimal counterexamples, extremal graph theory

**Mathematics Subject Classification:** 05C38, 05C35, 05C75

## 1 Introduction

The Erdős–Gyárfás conjecture asks whether every graph with minimum degree at least 3 contains a cycle whose length is a power of two [1]. The conjecture remains open in general and has been verified for several restricted classes, including 3-connected cubic planar graphs,  $P_8$ -free graphs, and  $P_{10}$ -free graphs [3, 4, 5]; a recent computer-aided preprint extends this to  $P_{13}$ -free graphs [6]. In a different direction, Liu and Montgomery [7] proved that every graph of sufficiently large average degree contains a cycle whose length is a power of two, answering a 1984 question of Erdős; the conjecture asks whether minimum degree 3 already suffices. Markström [2] showed by computer search that any cubic counterexample must contain at least 30 vertices.

All graphs considered here are finite, simple, and undirected. For a graph  $G$  we write  $V(G)$  and  $E(G)$  for its vertex and edge sets,  $d(v)$  for the degree of a vertex  $v$ ,  $N(v)$  for its neighborhood, and  $\delta(G)$  for the minimum degree. A graph  $H$  is a proper subgraph of  $G$  if  $H \subsetneq G$ . Following [8], a *minimal counterexample* to the Erdős–Gyárfás conjecture is a graph  $G$  with  $\delta(G) \geq 3$  containing no cycle whose length is a power of 2, chosen with minimum possible order and, subject to that, minimum possible size. Throughout, set

$$V_3 = \{v \in V(G) : d(v) = 3\}, \quad V_{\geq 4} = \{v \in V(G) : d(v) \geq 4\}.$$

Carr [8], building on an observation of Markström [2], proved that  $V_{\geq 4}$  is an independent set and that every vertex of a minimal counterexample is adjacent to a vertex of degree exactly 3,

and deduced that  $|V_3| \geq \frac{4}{7}|V(G)|$ . The proof of the  $4/7$  bound in [8] uses only the independence of  $V_{\geq 4}$ . Our first observation is that applying the domination property to the cubic vertices themselves improves the bound to  $2/3$  with no further machinery. Our second result uses deletions of vertex pairs, rather than the single-vertex and single-edge deletions of [8], to obtain new local structure around adjacent cubic vertices.

## 2 Preliminaries

We include short proofs of the following known facts to keep this note self-contained. Lemma 1 and Lemma 2 appear in [8]; Lemma 2(i) was first observed by Markström [2]. Since  $\delta(G) \geq 3$ , the graph  $G$  has at least four vertices, so every subgraph obtained from  $G$  by deleting at most two vertices is nonempty.

**Lemma 1** ([8]). *Let  $G$  be a minimal counterexample to the Erdős–Gyárfás conjecture. Then  $\delta(H) \leq 2$  for every nonempty proper subgraph  $H \subsetneq G$ .*

*Proof.* If  $H \subsetneq G$  satisfied  $\delta(H) \geq 3$ , then by minimality of  $G$  the graph  $H$  is not a counterexample, so  $H$  contains a cycle whose length is a power of 2. This cycle lies in  $G$ , a contradiction.  $\square$

**Lemma 2** ([2, 8]). *Let  $G$  be a minimal counterexample to the Erdős–Gyárfás conjecture.*

(i)  $V_{\geq 4}$  is an independent set.

(ii) Every vertex of  $G$  is adjacent to a vertex of degree exactly 3.

*Proof.* (i) If  $u, w \in V_{\geq 4}$  were adjacent, then deleting the edge  $uw$  leaves both endpoints with degree at least 3 and all other degrees unchanged, so  $\delta(G - uw) \geq 3$ , contradicting Lemma 1.

(ii) Let  $v \in V(G)$ . Since  $G - v$  is a nonempty proper subgraph, Lemma 1 gives  $\delta(G - v) \leq 2$ . Every vertex  $x \neq v$  satisfies  $d_{G-v}(x) \geq d(x) - 1 \geq 2$ , with  $d_{G-v}(x) \leq 2$  only if  $x \in N(v)$  and  $d(x) = 3$ . Hence  $v$  has a neighbor of degree exactly 3.  $\square$

## 3 Main results

**Theorem 3.** *Let  $G$  be a minimal counterexample to the Erdős–Gyárfás conjecture. Then*

$$|V_3| \geq \frac{2}{3}|V(G)|.$$

*Proof.* Let  $v \in V_3$ . By Lemma 2(ii),  $v$  is adjacent to a vertex of degree exactly 3, which lies in  $V_3$ . Since  $d(v) = 3$ , at most two of the edges incident with  $v$  join  $v$  to  $V_{\geq 4}$ . Summing over  $V_3$ ,

$$e(V_3, V_{\geq 4}) \leq 2|V_3|.$$

By Lemma 2(i), every neighbor of a vertex in  $V_{\geq 4}$  lies in  $V_3$ , so

$$e(V_3, V_{\geq 4}) = \sum_{u \in V_{\geq 4}} d(u) \geq 4|V_{\geq 4}|.$$

Combining the two displays gives  $|V_{\geq 4}| \leq \frac{1}{2}|V_3|$ , and since  $V_3$  and  $V_{\geq 4}$  partition  $V(G)$ ,

$$|V(G)| = |V_3| + |V_{\geq 4}| \leq \frac{3}{2}|V_3|,$$

which is the claimed bound.  $\square$

The next result uses deletions of two vertices at once. Note that since 4 is a power of 2, a minimal counterexample contains no 4-cycle; in particular, two adjacent vertices have at most one common neighbor.

**Proposition 4.** *Let  $G$  be a minimal counterexample to the Erdős–Gyárfás conjecture, and let  $vv' \in E(G)$  with  $d(v) = d(v') = 3$  and*

$$(N(v) \setminus \{v'\}) \cup (N(v') \setminus \{v\}) \subseteq V_{\geq 4}.$$

*Then  $v$  and  $v'$  have exactly one common neighbor  $u$ , and  $d(u) = 4$ . In particular,  $v$ ,  $v'$ , and  $u$  form a triangle whose apex  $u$  has degree exactly 4.*

*Proof.* Write  $N(v) = \{v', a, b\}$  and  $N(v') = \{v, a', b'\}$ , where  $a, b, a', b' \in V_{\geq 4}$  by hypothesis. Consider the nonempty proper subgraph  $H = G - \{v, v'\}$ . For any vertex  $x \notin \{v, v'\}$ ,

$$d_H(x) = d(x) - |N(x) \cap \{v, v'\}|.$$

A vertex  $x \notin \{v, v'\}$  is adjacent to  $v$  only if  $x \in \{a, b\}$ , and to  $v'$  only if  $x \in \{a', b'\}$ . If  $x$  is adjacent to neither  $v$  nor  $v'$ , then  $d_H(x) = d(x) \geq 3$ . If  $x$  is adjacent to exactly one of them, then  $x \in V_{\geq 4}$  and  $d_H(x) = d(x) - 1 \geq 3$ . By Lemma 1,  $\delta(H) \leq 2$ , so some  $x$  is adjacent to both  $v$  and  $v'$ , and then  $x \in \{a, b\} \cap \{a', b'\} \subseteq V_{\geq 4}$  with  $d(x) - 2 \leq 2$ , forcing  $d(x) = 4$ . Such an  $x$  is a common neighbor of  $v$  and  $v'$ . Since  $v$  and  $v'$  are adjacent and  $G$  contains no 4-cycle, this common neighbor is unique.  $\square$

*Remark 5.* Equality in Theorem 3 requires equality in both counting steps, so in any extremal configuration every vertex of  $V_{\geq 4}$  has degree exactly 4, and every cubic vertex has exactly one neighbor in  $V_3$ , so that  $V_3$  induces a perfect matching. Every matching edge then satisfies the hypothesis of Proposition 4, and hence lies in a triangle with an apex of degree exactly 4; moreover, the two remaining neighbors of the matched pair are distinct, since otherwise the pair would have two common neighbors and  $G$  would contain a 4-cycle. Each vertex of degree 4 can serve as the apex for at most two matching edges. Since the extremal configuration contains exactly  $|V_3|/2$  matching edges, each with a unique apex, and exactly  $|V_{\geq 4}| = |V_3|/2$  vertices of degree 4, the average number of matching edges served per degree-4 vertex is exactly 1; as this number lies in  $\{0, 1, 2\}$ , the number of degree-4 vertices serving two matching edges equals the number serving none. Sharpening the constant  $2/3$  would require ruling out, or further constraining, this rigid family of configurations, as well as the near-extremal configurations that approach it; the absence of 8-cycles appears to be the natural next tool.

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